

```
-- =====
-- STUDENT MANAGEMENT SYSTEM (ENTERPRISE VERSION)
-- DATABASE ARCHITECTURE UPGRADE
-- =====
```

```
DROP DATABASE IF EXISTS student_management_system;
CREATE DATABASE student_management_system;
USE student_management_system;
```

```
-- =====
-- 1. DEPARTMENTS
-- =====
```

```
CREATE TABLE departments (
    department_id INT PRIMARY KEY AUTO_INCREMENT,
    department_name VARCHAR(100) NOT NULL UNIQUE,
    department_code VARCHAR(20) NOT NULL UNIQUE,
    hod_name VARCHAR(100),
    office_phone VARCHAR(30),
    budget DECIMAL(15,2),
    established_date DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    ON UPDATE CURRENT_TIMESTAMP
);
```

```
-- =====
-- 2. SEMESTERS
-- =====
```

```
CREATE TABLE semesters (
    semester_id INT PRIMARY KEY AUTO_INCREMENT,
    semester_name VARCHAR(50) NOT NULL,
    academic_year VARCHAR(20) NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    status ENUM('Upcoming','Active','Completed')
    DEFAULT 'Upcoming',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

```
-- =====
-- 3. ROLES
-- =====
```

```
CREATE TABLE roles (
    role_id INT PRIMARY KEY AUTO_INCREMENT,
    role_name VARCHAR(50) UNIQUE NOT NULL,
    description TEXT
);
```

```
-- =====
-- 4. PERMISSIONS
-- =====
```

```
CREATE TABLE permissions (
    permission_id INT PRIMARY KEY AUTO_INCREMENT,
    permission_name VARCHAR(100) UNIQUE NOT NULL,
    module_name VARCHAR(100),
    description TEXT
);
```

```
-- =====
-- 5. ROLE PERMISSIONS
-- =====
```

```
CREATE TABLE role_permissions (
    role_permission_id INT PRIMARY KEY AUTO_INCREMENT,
    role_id INT NOT NULL,
    permission_id INT NOT NULL,

    FOREIGN KEY (role_id)
        REFERENCES roles(role_id)
        ON DELETE CASCADE,

    FOREIGN KEY (permission_id)
        REFERENCES permissions(permission_id)
        ON DELETE CASCADE,

    UNIQUE(role_id, permission_id)
);
```

```
-- =====
-- 6. STUDENTS
-- =====
```

```
CREATE TABLE students (
    student_id INT PRIMARY KEY AUTO_INCREMENT,
    student_number VARCHAR(30) UNIQUE NOT NULL,

    first_name VARCHAR(50) NOT NULL,
    middle_name VARCHAR(50),
    last_name VARCHAR(50) NOT NULL,

    gender ENUM('Male', 'Female', 'Other') NOT NULL,

    date_of_birth DATE NOT NULL,

    email VARCHAR(100) UNIQUE NOT NULL,
    phone VARCHAR(30),
```

```

address TEXT,
city VARCHAR(50),
state VARCHAR(50),
postal_code VARCHAR(20),
country VARCHAR(50),

photo VARCHAR(255),

department_id INT,

admission_date DATE NOT NULL,
graduation_date DATE,

gpa DECIMAL(3,2) DEFAULT 0.00,

status ENUM(
    'Active',
    'Inactive',
    'Graduated',
    'Suspended',
    'Withdrawn'
) DEFAULT 'Active',

deleted_at TIMESTAMP NULL,

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    ON UPDATE CURRENT_TIMESTAMP,

FOREIGN KEY (department_id)
    REFERENCES departments(department_id)
    ON DELETE SET NULL
);

-- =====
-- 7. STUDENT GUARDIANS
-- =====

CREATE TABLE student_guardians (
    guardian_id INT PRIMARY KEY AUTO_INCREMENT,

    student_id INT NOT NULL,

    full_name VARCHAR(100) NOT NULL,
    relationship VARCHAR(50),
    phone VARCHAR(30),
    email VARCHAR(100),
    address TEXT,

```

```

        created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

        FOREIGN KEY (student_id)
            REFERENCES students(student_id)
            ON DELETE CASCADE
    );

-- =====
-- 8. DOCUMENTS
-- =====

CREATE TABLE documents (
    document_id INT PRIMARY KEY AUTO_INCREMENT,

    student_id INT NOT NULL,

    document_type VARCHAR(100),
    file_name VARCHAR(255),
    file_path VARCHAR(255),

    uploaded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (student_id)
        REFERENCES students(student_id)
        ON DELETE CASCADE
);

-- =====
-- 9. INSTRUCTORS
-- =====

CREATE TABLE instructors (
    instructor_id INT PRIMARY KEY AUTO_INCREMENT,

    instructor_number VARCHAR(30) UNIQUE NOT NULL,

    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,

    gender ENUM('Male', 'Female', 'Other'),

    email VARCHAR(100) UNIQUE NOT NULL,
    phone VARCHAR(30),

    specialization VARCHAR(100),

    department_id INT,

    hire_date DATE NOT NULL,

```

```

salary DECIMAL(12,2),

status ENUM(
    'Active',
    'On Leave',
    'Retired',
    'Inactive'
) DEFAULT 'Active',

photo VARCHAR(255),

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

FOREIGN KEY (department_id)
    REFERENCES departments(department_id)
    ON DELETE SET NULL
);

```

```

-- =====
-- 10. COURSES
-- =====

```

```

CREATE TABLE courses (
    course_id INT PRIMARY KEY AUTO_INCREMENT,

    course_code VARCHAR(20) UNIQUE NOT NULL,
    course_name VARCHAR(100) NOT NULL,

    credits INT NOT NULL,

    description TEXT,

    department_id INT,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (department_id)
        REFERENCES departments(department_id)
        ON DELETE SET NULL
);

```

```

-- =====
-- 11. COURSE OFFERINGS
-- =====

```

```

CREATE TABLE course_offerings (
    offering_id INT PRIMARY KEY AUTO_INCREMENT,

    course_id INT NOT NULL,
    instructor_id INT,

```

```

semester_id INT NOT NULL,

max_students INT DEFAULT 50,

room VARCHAR(50),

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

FOREIGN KEY (course_id)
    REFERENCES courses(course_id)
    ON DELETE CASCADE,

FOREIGN KEY (instructor_id)
    REFERENCES instructors(instructor_id)
    ON DELETE SET NULL,

FOREIGN KEY (semester_id)
    REFERENCES semesters(semester_id)
    ON DELETE CASCADE
);

```

```

-- =====
-- 12. CLASSROOMS
-- =====

```

```

CREATE TABLE classrooms (
    classroom_id INT PRIMARY KEY AUTO_INCREMENT,

    room_name VARCHAR(50),
    building_name VARCHAR(100),
    capacity INT,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

```

-- =====
-- 13. SCHEDULES / TIMETABLES
-- =====

```

```

CREATE TABLE schedules (
    schedule_id INT PRIMARY KEY AUTO_INCREMENT,

    offering_id INT NOT NULL,
    classroom_id INT,

    day_of_week ENUM(
        'Monday',
        'Tuesday',
        'Wednesday',
        'Thursday',

```

```

        'Friday',
        'Saturday',
        'Sunday'
    ),

    start_time TIME,
    end_time TIME,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (offering_id)
        REFERENCES course_offerings(offering_id)
        ON DELETE CASCADE,

    FOREIGN KEY (classroom_id)
        REFERENCES classrooms(classroom_id)
        ON DELETE SET NULL
);

-- =====
-- 14. ENROLLMENTS
-- =====

CREATE TABLE enrollments (
    enrollment_id INT PRIMARY KEY AUTO_INCREMENT,

    student_id INT NOT NULL,
    offering_id INT NOT NULL,

    enrollment_date DATE NOT NULL,

    status ENUM(
        'Enrolled',
        'Completed',
        'Dropped',
        'Incomplete'
    ) DEFAULT 'Enrolled',

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (student_id)
        REFERENCES students(student_id)
        ON DELETE CASCADE,

    FOREIGN KEY (offering_id)
        REFERENCES course_offerings(offering_id)
        ON DELETE CASCADE,

    UNIQUE(student_id, offering_id)
);

```

```
-- =====
-- 15. RESULTS
-- =====
```

```
CREATE TABLE results (
    result_id INT PRIMARY KEY AUTO_INCREMENT,

    enrollment_id INT NOT NULL,

    score DECIMAL(5,2),

    grade_letter VARCHAR(2),

    grade_point DECIMAL(3,2),

    remarks TEXT,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (enrollment_id)
        REFERENCES enrollments(enrollment_id)
        ON DELETE CASCADE
);
```

```
-- =====
-- 16. ATTENDANCE SESSIONS
-- =====
```

```
CREATE TABLE attendance_sessions (
    session_id INT PRIMARY KEY AUTO_INCREMENT,

    offering_id INT NOT NULL,

    session_topic VARCHAR(255),

    session_date DATE NOT NULL,

    start_time TIME,
    end_time TIME,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (offering_id)
        REFERENCES course_offerings(offering_id)
        ON DELETE CASCADE
);
```

```
-- =====
-- 17. ATTENDANCE RECORDS
```



```
-- =====
```

```
CREATE TABLE attendance_records (  
    attendance_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    session_id INT NOT NULL,  
    student_id INT NOT NULL,  
  
    attendance_status ENUM(  
        'Present',  
        'Absent',  
        'Late',  
        'Excused'  
    ) NOT NULL,  
  
    remarks TEXT,  
  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
    FOREIGN KEY (session_id)  
        REFERENCES attendance_sessions(session_id)  
        ON DELETE CASCADE,  
  
    FOREIGN KEY (student_id)  
        REFERENCES students(student_id)  
        ON DELETE CASCADE,  
  
    UNIQUE(session_id, student_id)  
);
```

```
-- =====
```

```
-- 18. FEE STRUCTURES
```

```
-- =====
```

```
CREATE TABLE fee_structures (  
    fee_structure_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    department_id INT,  
    semester_id INT,  
  
    fee_type ENUM(  
        'Tuition',  
        'Library',  
        'Lab',  
        'Hostel',  
        'Examination',  
        'Other'  
    ),  
  
    amount DECIMAL(12,2),
```

```

        created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

        FOREIGN KEY (department_id)
            REFERENCES departments(department_id)
            ON DELETE SET NULL,

        FOREIGN KEY (semester_id)
            REFERENCES semesters(semester_id)
            ON DELETE SET NULL
    );

-- =====
-- 19. INVOICES
-- =====

CREATE TABLE invoices (
    invoice_id INT PRIMARY KEY AUTO_INCREMENT,

    student_id INT NOT NULL,
    semester_id INT,

    invoice_number VARCHAR(50) UNIQUE,

    total_amount DECIMAL(12,2),
    amount_paid DECIMAL(12,2) DEFAULT 0,
    balance DECIMAL(12,2),

    status ENUM(
        'Pending',
        'Partially Paid',
        'Paid',
        'Cancelled'
    ) DEFAULT 'Pending',

    due_date DATE,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (student_id)
        REFERENCES students(student_id)
        ON DELETE CASCADE,

    FOREIGN KEY (semester_id)
        REFERENCES semesters(semester_id)
        ON DELETE SET NULL
);

-- =====
-- 20. INVOICE ITEMS

```

```
-- =====
```

```
CREATE TABLE invoice_items (  
    invoice_item_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    invoice_id INT NOT NULL,  
  
    fee_type VARCHAR(100),  
    amount DECIMAL(12,2),  
  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
    FOREIGN KEY (invoice_id)  
        REFERENCES invoices(invoice_id)  
        ON DELETE CASCADE  
);
```

```
-- =====
```

```
-- 21. PAYMENTS
```

```
-- =====
```

```
CREATE TABLE payments (  
    payment_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    invoice_id INT NOT NULL,  
  
    amount_paid DECIMAL(12,2) NOT NULL,  
  
    payment_date DATE NOT NULL,  
  
    payment_method ENUM(  
        'Cash',  
        'Bank Transfer',  
        'Credit Card',  
        'Debit Card',  
        'Mobile Money',  
        'Cheque'  
    ),  
  
    transaction_reference VARCHAR(100),  
  
    payment_status ENUM(  
        'Pending',  
        'Completed',  
        'Failed',  
        'Refunded'  
    ) DEFAULT 'Completed',  
  
    remarks TEXT,
```

```

        created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

        FOREIGN KEY (invoice_id)
            REFERENCES invoices(invoice_id)
            ON DELETE CASCADE
    );

-- =====
-- 22. SCHOLARSHIPS
-- =====

CREATE TABLE scholarships (
    scholarship_id INT PRIMARY KEY AUTO_INCREMENT,

    student_id INT NOT NULL,

    scholarship_name VARCHAR(100),
    sponsor_name VARCHAR(100),

    amount DECIMAL(12,2),

    start_date DATE,
    end_date DATE,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (student_id)
        REFERENCES students(student_id)
        ON DELETE CASCADE
);

-- =====
-- 23. BOOKS
-- =====

CREATE TABLE books (
    book_id INT PRIMARY KEY AUTO_INCREMENT,

    isbn VARCHAR(30) UNIQUE,

    title VARCHAR(255),
    author VARCHAR(255),
    publisher VARCHAR(255),

    publication_year YEAR,

    total_copies INT DEFAULT 1,
    available_copies INT DEFAULT 1,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

```

```
);
```

```
-- =====  
-- 24. BORROWED BOOKS  
-- =====
```

```
CREATE TABLE borrowed_books (  
    borrow_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    student_id INT NOT NULL,  
    book_id INT NOT NULL,  
  
    issue_date DATE NOT NULL,  
    due_date DATE NOT NULL,  
    return_date DATE,  
  
    fine_amount DECIMAL(10,2) DEFAULT 0,  
  
    status ENUM(  
        'Issued',  
        'Returned',  
        'Overdue',  
        'Lost'  
    ) DEFAULT 'Issued',  
  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
    FOREIGN KEY (student_id)  
        REFERENCES students(student_id)  
        ON DELETE CASCADE,  
  
    FOREIGN KEY (book_id)  
        REFERENCES books(book_id)  
        ON DELETE CASCADE  
);
```

```
-- =====  
-- 25. NOTIFICATIONS  
-- =====
```

```
CREATE TABLE notifications (  
    notification_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    title VARCHAR(255),  
    message TEXT,  
  
    notification_type ENUM(  
        'General',  
        'Academic',  
        'Finance',
```

```

        'Attendance',
        'System'
    ),

    created_by INT,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- 26. USER NOTIFICATIONS
-- =====

CREATE TABLE user_notifications (
    user_notification_id INT PRIMARY KEY AUTO_INCREMENT,

    notification_id INT NOT NULL,
    user_id INT NOT NULL,

    is_read BOOLEAN DEFAULT FALSE,
    read_at TIMESTAMP NULL,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- =====
-- 27. USERS
-- =====

CREATE TABLE users (
    user_id INT PRIMARY KEY AUTO_INCREMENT,

    username VARCHAR(50) UNIQUE NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,

    password_hash VARCHAR(255) NOT NULL,

    role_id INT NOT NULL,

    student_id INT NULL,
    instructor_id INT NULL,

    is_active BOOLEAN DEFAULT TRUE,

    last_login TIMESTAMP NULL,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (role_id)
        REFERENCES roles(role_id)

```

```

        ON DELETE RESTRICT,

FOREIGN KEY (student_id)
    REFERENCES students(student_id)
    ON DELETE CASCADE,

FOREIGN KEY (instructor_id)
    REFERENCES instructors(instructor_id)
    ON DELETE CASCADE
);

CREATE TABLE password_resets (
    reset_id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT,
    reset_token VARCHAR(255),
    expires_at DATETIME,
    used BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (user_id)
        REFERENCES users(user_id)
        ON DELETE CASCADE
);

-- =====
-- 28. AUDIT LOGS
-- =====

CREATE TABLE audit_logs (
    log_id INT PRIMARY KEY AUTO_INCREMENT,

    user_id INT,

    action_type VARCHAR(100),
    table_name VARCHAR(100),
    record_id INT,

    old_values JSON,
    new_values JSON,

    ip_address VARCHAR(100),

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (user_id)
        REFERENCES users(user_id)
        ON DELETE SET NULL
);

-- =====
-- 29. PAYROLL

```

```
-- =====
```

```
CREATE TABLE payroll (  
    payroll_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    instructor_id INT NOT NULL,  
  
    basic_salary DECIMAL(12,2),  
    allowances DECIMAL(12,2),  
    deductions DECIMAL(12,2),  
  
    net_salary DECIMAL(12,2),  
  
    payment_month VARCHAR(20),  
    payment_year YEAR,  
  
    payment_status ENUM(  
        'Pending',  
        'Paid'  
    ) DEFAULT 'Pending',  
  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
    FOREIGN KEY (instructor_id)  
        REFERENCES instructors(instructor_id)  
        ON DELETE CASCADE  
);
```

```
-- =====
```

```
-- 30. SYSTEM SETTINGS
```

```
-- =====
```

```
CREATE TABLE system_settings (  
    setting_id INT PRIMARY KEY AUTO_INCREMENT,  
  
    setting_key VARCHAR(100) UNIQUE,  
    setting_value TEXT,  
  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
-- =====
```

```
-- INDEXES
```

```
-- =====
```

```
CREATE INDEX idx_students_department  
ON students(department_id);
```

```
CREATE INDEX idx_students_status  
ON students(status);
```



```
CREATE INDEX idx_enrollments_student
ON enrollments(student_id);
```

```
CREATE INDEX idx_enrollments_offering
ON enrollments(offering_id);
```

```
CREATE INDEX idx_results_enrollment
ON results(enrollment_id);
```

```
CREATE INDEX idx_attendance_session
ON attendance_records(session_id);
```

```
CREATE INDEX idx_payments_invoice
ON payments(invoice_id);
```

```
CREATE INDEX idx_users_role
ON users(role_id);
```

```
CREATE INDEX idx_notifications_type
ON notifications(notification_type);
```

```
-- =====
-- SAMPLE ROLES
-- =====
```

```
INSERT INTO roles (role_name, description)
VALUES
('Super Admin', 'Full system access'),
('Admin', 'Administrative operations'),
('Instructor', 'Manage classes and results'),
('Student', 'Student portal access'),
('Accountant', 'Manage financial records');
```

```
-- =====
-- SAMPLE PERMISSIONS
-- =====
```

```
INSERT INTO permissions (permission_name, module_name)
VALUES
('manage_students', 'Students'),
('manage_courses', 'Courses'),
('manage_payments', 'Finance'),
('manage_attendance', 'Attendance'),
('manage_users', 'Administration');
```

```
-- =====
-- DATABASE COMPLETE
-- =====
```

```
SELECT 'Student Management System Database Created Successfully'  
AS STATUS;
```